Corollary 5.19.2. *As a representation of $S_n \times GL(V)$, $V^{\otimes n}$ decomposes as $\bigoplus_\lambda V_\lambda \otimes L_\lambda$, where $L_\lambda = \mathrm{Hom}_{S_n}(V_\lambda, V^{\otimes n})$ are distinct irreducible representations of $GL(V)$ or zero.*

Example 5.19.3. If $\lambda = (n)$, then $L_\lambda = S^n V$, and if $\lambda = (1^n)$ (n copies of 1), then $L_\lambda = \wedge^n V$. It was shown in Problem 4.12.3 that these representations are indeed irreducible (except that $\wedge^n V$ is zero if $n > \dim V$).

5.20. Historical interlude: Hermann Weyl at the intersection of limitation and freedom

Hermann Weyl (1885–1955) received his doctorate at the University of Göttingen under the guidance of David Hilbert, whom he later paid a rather dubious compliment by calling him "the Pied Piper ... seducing so many rats to follow him into the deep river of mathematics". Hilbert called mathematical physics "a vital nerve" of mathematics, and Weyl inherited the interest in cross-fertilization of mathematics and physics from his teacher (quoted in [**53**, pp. 357, 358]). The willingness of Göttingen mathematicians to get their formulas dirty by engaging physical problems set them apart from the obsessive purism of the Berlin mathematical school.

In 1913 Weyl was offered a professorship at the ETH in Zürich, despite a somewhat lukewarm endorsement from Frobenius, who regarded the work of all candidates who came from the Göttingen school as "very general, very deep, so abysmally deep that a shortsighted person like me finds it difficult to recognize new ideas" (quoted in [**22**, p. 421]). Weyl accepted and found himself in the company of Albert Einstein, who was teaching at the ETH at the time. Yet it took a world war to make Weyl pay attention to what Einstein was doing. The German government initially judged Weyl unfit to fight in World War I, but as losses were mounting, it reevaluated the concept of physical fitness and dragged him onto the battlefield. Weyl's encounter with the German army brought little satisfaction to either side. In 1916, at the request of the Swiss government, Weyl was discharged and allowed to resume his work at the ETH. "My mathematical mind was as blank as any veteran's, and I did not know what to do", he later recalled. "I began to study algebraic surfaces; but before I had